

EXAM 7 – SPRING 2013

5. (2.25 points)

Given the following information as of December 31, 2012:

	<u>Cumulative Reported Loss (\$000)</u>			
<u>Accident Year</u>	<u>12</u>	<u>24</u>	<u>36</u>	<u>48</u>
2009	233	522	570	601
2010	204	480	512	
2011	242	541		
2012	255			

- Claim data in the triangle is on an unlimited basis
- Basic limit: \$50,000
- Calendar period ground-up unlimited loss trend: 5% per year
- An exponential distribution fit to the claim level data at each maturity produced the following unlimited claim size means at the accident year 2012 level:

	<u>12</u>	<u>24</u>	<u>36</u>	<u>48</u>
Unlimited Claim Size Mean	26,000	48,000	58,000	66,000

- Mean of exponential distribution: θ
- Variance of exponential distribution: θ^2
- Limited mean of exponential distribution at limit k : $\theta \left(1 - e^{-\frac{k}{\theta}}\right)$

Using the method described by Sahasrabudhe, calculate the latest diagonal of the cumulative reported loss triangle based on accident year 2012 cost levels for the basic limit of loss.

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Question 5 Sample Answer

Solution 1

Triangle of Trend

<u>AY</u>	<u>12</u>	<u>24</u>	<u>36</u>	<u>48</u>
09	1	1.05	1.103	1.158
10	1.05	1.103	1.158	1.216
11	1.103	1.158	1.216	1.276
12	1.158	1.216	1.276	1.34

Triangle of Trended Θ 's

<u>AY</u>	<u>12</u>	<u>24</u>	<u>36</u>	<u>48</u>
09	22,453	41,447	50,136	57,036
10	23,575	43,539	52,636	
11	24,765	45,711		
12	26,000	48,000	58,000	66,000

AY 09 at 48 months = $66,000 \times (1.158/1.34)$

LEV's for basic limit – last row only

$$\begin{array}{cccc} \underline{12} & \underline{24} & \underline{36} & \underline{48} \\ 22,200 & 31,062 & 33,507 & 35,059 \\ \wedge & & & \\ 26,000(1-e^{-50,000/26,000}) & & & \end{array}$$

<u>AY</u>	<u>Age</u>	<u>cum loss @ B cost level</u>
2009	48	$601 \times 35,059 / 57,036 = 369.42$
2010	36	$512 \times 33,507 / 52,636 = 325.93$
2011	24	$541 \times 31,026 / 45,711 = 367.63$
2012	12	$255 \times 22,200 / 26,000 = 217.73$

Solution 2

Mean clm at BC at 2012 lvl

$$\begin{array}{cccc} 12 & 24 & 36 & 48 \\ 22,200 & 31,062 & 33,507 & 35,059 \\ 22,200 = 26,000 \times (1-e^{-50,000/26,000}) & & & \end{array}$$

For latest CY

$$\begin{aligned} 2012 \text{ Adjusted} &= 255 \times 22,200 / 26,000 = 218 \\ 2011 \text{ Adjusted} &= 541 \times 1.05 \times 31,062 / 48,000 = 368 \\ 2010 \text{ Adjusted} &= 512 \times 1.05^2 \times 33,507 / 58,000 = 326 \\ 2009 \text{ Adjusted} &= 601 \times 1.05^3 \times 35,509 / 66,000 = 370 \end{aligned}$$

Examiner Comment

The question was fairly straightforward, with a number of acceptable approaches to the same answer. There was a typo in the given exponential limited mean formula, but this was recognized and corrected by almost all of the candidates. The most common error (other than

the most basic of trending errors) was calculating and using the LEV's on the latest diagonal for each AY, as opposed to AY 2012 at the various evaluations. Following this line, some candidates knew how to calculate the individual pieces, but just couldn't wrap it all together.